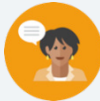


## 9.6 Describing and Interpreting Histograms – Part 1

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |                                                                                                                                                                                                             |                                                                                                                                     |  |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--|
|  Benchmark           | MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.                                                                                                                                               |                                                                                      |  Learning Target                                                                                                         | <ul style="list-style-type: none"><li>I can qualitatively describe and interpret a histogram within a real-world context.</li></ul> |  |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                                                                    |                                                                                      | Working Toward                                                                                                                                                                                              |                                                                                                                                     |  |
|                                                                                                       | MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.                                                                                                                                                                                                                         |                                                                                      | MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers. |                                                                                                                                     |  |
|  Essential Questions | <ul style="list-style-type: none"><li>What is the purpose of a histogram?</li><li>How do you read a histogram?</li><li>How can you use a histogram to analyze data?</li></ul>                                                                                                                                                                                                  |                                                                                      |                                                                                                                                                                                                             |                                                                                                                                     |  |
|  Lesson Narrative    | This lesson is Part 1 of 2 in which students interpret histograms. Students begin by formally defining what a histogram is and then use a histogram to answer questions about a data set. Students continue to interpret data displayed in histograms and analyze the spread of the data to determine typical values and where measures of center can be found on the display. |                                                                                      |                                                                                                                                                                                                             |                                                                                                                                     |  |
|  Component Summary | 9.6.1 Warm-Up (~5 mins)                                                                                                                                                                                                                                                                                                                                                        | Examine why statistics are important ( <i>SE p. 174</i> )                            | 9.6.4 Try It (5-10 mins)                                                                                                                                                                                    | Analyze a histogram ( <i>SE p. 177</i> )                                                                                            |  |
|                                                                                                       | 9.6.2 Guided Instruction (5-10 mins)                                                                                                                                                                                                                                                                                                                                           | Use a histogram to understand data ( <i>SE p. 174</i> )                              | 9.6.5 Your Turn! (~5 mins)                                                                                                                                                                                  | Interpret a histogram ( <i>SE p. 178</i> )                                                                                          |  |
|                                                                                                       | 9.6.3 Exploration (~10 mins)                                                                                                                                                                                                                                                                                                                                                   | Interpret the relationship between a data table and a histogram ( <i>SE p. 175</i> ) | 9.6.6 Wrap-Up (~5 mins)                                                                                                                                                                                     | Self-assessment of learning target ( <i>SE p. 178</i> )                                                                             |  |
|                                                                                                       | Glossary                                                                                                                                                                                                                                                                                                                                                                       | Materials                                                                            | Additional Preparation                                                                                                                                                                                      |                                                                                                                                     |  |
|  Resources         | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Histogram</li></ul> <u>Additional Terms:</u> N/A                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"><li>(Optional) Colored pencils</li></ul>           | <ul style="list-style-type: none"><li>9.6.1 Warm-Up contains a video to accompany the question prompt.</li></ul>                                                                                            |                                                                                                                                     |  |

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Things to Consider                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                                                                | <ul style="list-style-type: none"><li>Students may confuse histograms and bar graphs.</li><li>Students may struggle to understand the concept of frequency.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"><li>Consider using the same set of data to create a histogram and a bar graph. Encourage students to notice the differences and similarities in the graphical representations.</li><li>Have students pick a color: red, blue, yellow, or green. Determine how frequently each color is chosen as a way to demonstrate frequency to students.</li></ul> |
|                                                                                                                | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                       |
|                                                                                                                | <b>Notice and Wonder</b><br>9.6.2 Guided Instruction invites students to use a histogram to analyze a set of data. Consider applying Notice and Wonder routines to the histogram, encouraging students to discover the relationship between the data set and the representation of it in the histogram.                                                                                                                                                                                                                                                                                                                                                                             | <b>MA.K12.MTR.7.1: Real-World Contexts</b><br>Students will be working to represent and analyze real-world data in a histogram. Clarifications within this standard emphasize using models and methods to represent everyday experiences. Utilize this lesson to provide opportunities for students to use histograms to investigate data.                                               |
| <br>Differentiate Instruction | This lesson focuses heavily on the visual representation of data through histograms. Representing data sets graphically allows students to more easily make connections between statistical measures and the data set. Encourage students to color-code different bars in the histogram and draw lines to the corresponding frequency to provide additional visual and kinesthetic entry points.<br><br>Partner discussion and reading questions aloud provide auditory entry points for students to engage with the content as well. Consider points in the lesson when students who may be struggling could benefit from the added support of partner- or teacher-led discussion. |                                                                                                                                                                                                                                                                                                                                                                                          |
| Lesson Notes:                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                          |

## 9.6.1 Warm-Up: Why Statistics?

**Component Overview:** Have students watch the video before reading the description and answering the prompt. Students will then consider additional reasons why statistics are important and how they can be utilized in daily life.



### 9.6.1 Warm-Up: Why Statistics?

Eric Newburger relates the need for studying statistics to the need for using a microscope. He discusses how a microscope is used to study small things. It's also needed to study things like food poisoning at a larger level to aid in understanding diseases and how viruses spread.



In Mr. Newburger's work, he explains you can take a big thing and shrink it down small enough to study and to understand. That's the study of statistics. And when you want to understand something that is too small, you can use a tool (like the microscope), which is "why statistics."



1. List two reasons why statistics are important.

*Answers will vary. There is a need for statistics because it allows us to present data and analyze data. Statistics also allows us to make predictions based on trends found in the data.*



*Do you know someone who is a statistician or works in data management? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around statistics as a career.*

## 9.6.2 Guided Instruction: Histograms

**Component Overview:** Begin this section by reviewing the definition of a histogram as a class. Then have students work in pairs to analyze and interpret the data presented in the histogram about shark lengths.



### 9.6.2 Guided Instruction: Histograms

Numerical data can be displayed graphically in a variety of ways. One type of graph is a histogram.



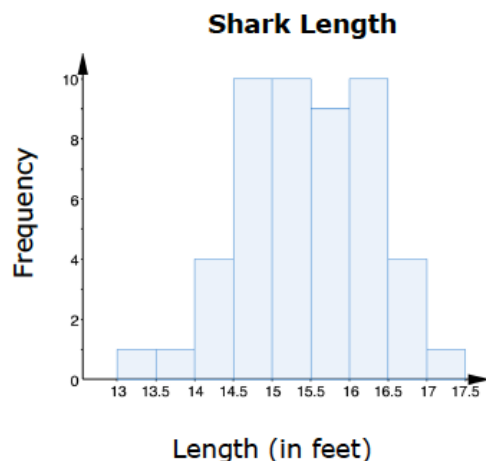
A **histogram** is a visual display of numerical data using bars along a number line with no spaces between the bars. The height of each bar represents either the frequency or relative frequency of data within that interval.



#### Notice and Wonder

Consider creating a T-chart on the board with the headings “Notice” and “Wonder.” Invite students to point out the observations they notice and to cite the details they wonder about regarding the histogram. Use students’ comments to complete the T-chart.

1. The histogram summarizes the lengths, in feet (ft), of a group of adult female sharks.



Discuss the term “frequency” with students. Have students pick a color: red, blue, yellow, or green. Determine how frequently each color is chosen as a way to demonstrate frequency to students. Color-code students’ responses on the board to demonstrate the frequency.

- a. With your partner, discuss the labels for the vertical and horizontal axes.

*Students do not need to write an explanation here, but a sample response is provided for reference. The horizontal axis represents the length of the female sharks. The vertical axis represents the number of sharks with lengths in the various intervals.*

## 9.6.2 Guided Instruction: Histograms (continued)

- b. How many sharks were between 15.5 ft and 16 ft?

*9 sharks*

- c. How many total sharks were measured to create the histogram?

*50 sharks*

2. Each bar of a histogram includes the left-end value, but not the right-end value. For example, the first bar includes sharks that are 13 ft and 13.25 ft, but not 13.5 ft.

- a. Discuss with your partner whether or not you can determine the exact lengths of each shark measured from the histogram. Summarize your discussion.

*Answers will vary. The exact length of the sharks cannot be determined because the histogram displays the number of (frequency) of sharks for each interval of lengths.*

- b. Discuss with your partner your observations about the width of each bar.

*Students do not need to write an explanation here, but a sample response is provided for reference. The width of the bars are equal.*

3. If male sharks were added to the sample, hypothesize what would happen to the spread of the data.

*Answers will vary. Male sharks are typically smaller than female sharks, so adding the lengths of the male sharks will likely not change the range of the data.*



To draw attention to the way in which histograms differ from other graphical representations and present data, ask questions like:

- How does each bar of the histogram show the range of lengths of sharks? How does this differ from showing the exact lengths of sharks?
- What is the mode of the data? Can you determine it using the histogram? Why or why not?
- Where do you think the median would be?

## 9.6.3 Exploration: Histograms

**Component Overview:** In this activity, students notice how the data in the table translates to the histogram. Students may work independently or with a partner to complete the activity. Partners may work together on all of the questions or divide the items and then share their answers with their partner.

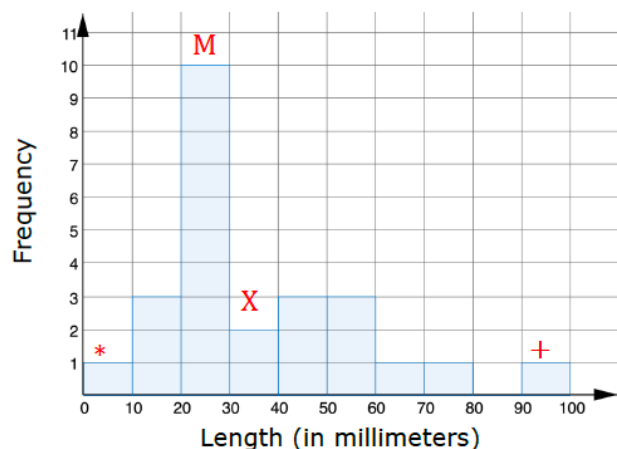


### 9.6.3 Exploration: Histograms

An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer recorded the earthworms' lengths, in millimeters, in a table and then created a histogram to display the data.

| Earthworm Lengths in Millimeters (mm) |    |    |    |    |    |    |    |    |    |
|---------------------------------------|----|----|----|----|----|----|----|----|----|
| 6                                     | 11 | 18 | 19 | 20 | 23 | 23 | 25 | 25 | 26 |
| 27                                    | 27 | 28 | 29 | 32 | 33 | 41 | 42 | 48 | 52 |
| 54                                    | 59 | 60 | 77 | 93 |    |    |    |    |    |



Some students may struggle to track the data with their eyes. Encourage students to color-code different bars in the histogram and to draw a horizontal line from the top of each bar to its corresponding frequency on the y-axis.

1. How many earthworms were measured in the data?

25

2. The number of earthworms measured can be

determined from

- ☐ the data set.
- ☐ the histogram.
- ☒ the data set and the histogram.

### 9.6.3 Exploration: Histograms (continued)

3. The length of the shortest earthworm can be

determined from

- ☐ the data set.
- ☐ the histogram.
- ☐ the data set and the histogram.

4. Place a "\*" above the bar that contains the value for the shortest earthworm.

*The smallest data value is in the first column.*

5. Determine the length of the longest earthworm. Then, place a "+" above the bar that contains the value for the longest earthworm.

*The longest earthworm was 93 mm and is located in the last column.*

6. The mean length of the earthworms can be determined

from

- ☐ the data set.
- ☐ the histogram.
- ☐ the data set and the histogram.

7. What is the mean length of the earthworms?

*35.92*

8. Half of the earthworms are longer than 28 millimeters.

9. Mark an "X" above the bar that contains the mean and mark an "M" above the bar that contains the median.

*Added to original histogram*

10. Discuss with your partner how the histogram would change if a 100-millimeter long earthworm was added to the data set. Summarize your discussion.

*Answers will vary. Adding a 100-millimeter long earthworm to the data set will add another bar to the histogram, since that earthworm would be included in the interval 100 – 110.*



To help students discern the differences and make connections between the different ways to display data, ask questions like:

- How does the data set help you visually understand the data?
- Do you think the data is easier to analyze listed individually or grouped in the histogram? Why?
- What are the strengths of just using the data set?
- What are the strengths of just using the histogram?



Challenge students to determine the mode and the range of the data. If students feel comfortable with box plots, have them create a box plot from the data.



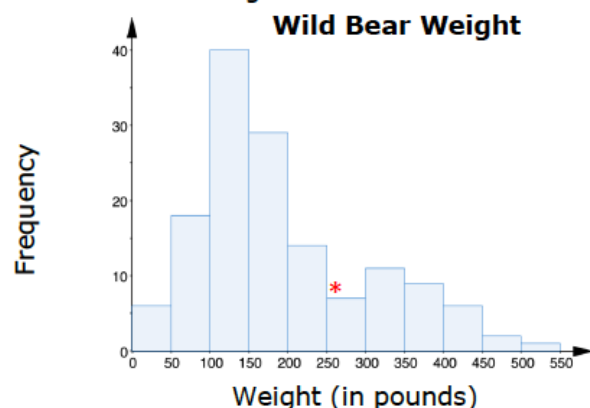
## 9.6.4 Try It: Analyzing Histograms

**Component Overview:** Students use a histogram to analyze the data contained in it. Some students will feel comfortable completing this activity independently, while other students will feel more comfortable with support.



### 9.6.4 Try It: Analyzing Histograms

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



- a. Could this sample contain a bear weighing 250 lbs? Explain.  
*Yes, the 250 pounds bear will fall between the 250 and 300 weight range.*
- b. Mark a "\*" above that bar that would contain a 250-lb bear. *Shown on histogram.*
- c. Based on the graph, what can be concluded about the weight of the heaviest bear?  
*The heaviest bear is in the last interval of weights, 500 to 550, but does not include 550.*
- d. What is a typical weight for the bears in this graph? Include units.  
*Answers will vary. Between 150 and 200 pounds.*
- e. Do more than half of the bears in the study weigh less than 250 lbs?  
*Yes, more than half of the bears weigh less than 250 pounds.*



Invite students to share any initial impressions they have of the histogram regarding the distribution of the data. Remind students to read the introductory paragraph and think about any suppositions they have about the relation of a bear's age to its weight.



Part 2 of this lesson will focus on student understanding of the shape of histograms and how to detect any gaps, clusters, or outliers in the data set. Position students to begin to explicitly analyze the distribution of data:

- How would you describe the shape of this histogram? Does it look the same on either side of the median? Why or why not?
- How can you use the histogram to determine a "typical" weight?
- Do any of the data points on the histogram seem out of place? Why?



## 9.6.5 Your Turn!

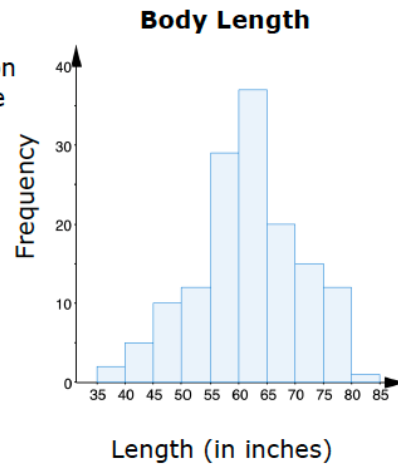
**Component Overview:** In this activity, students will synthesize what they have learned about analyzing a histogram. Students may complete the activity independently, with a partner, or in a small group.



### 9.6.5 Your Turn!

1. The histogram summarizes the data on the body lengths of the same 143 wild bears.

- a. Does this sample contain a bear that is 85 inches long? Explain your reasoning.



*No, this sample does not contain a bear that is 85 inches long. The largest bear is between 80 and just up to 85 inches.*

- b. What is a typical length for the bears in this sample? Include units.  
*Between 60 and 65 inches.*
- c. Is the spread larger for the weight of bears or the length of bears?  
*The spread is larger for the weight of bears.*



Students may struggle to determine the typical length of the bears. Ask questions to further scaffold this section:

- How can you determine which length is typical using a histogram? How can you use the visual display to determine a typical weight?
- How do you determine frequency values between the marked numbers on the y-axis?
- What is the range of the bears' body lengths?

## 9.6.6 Wrap-Up: Reflection

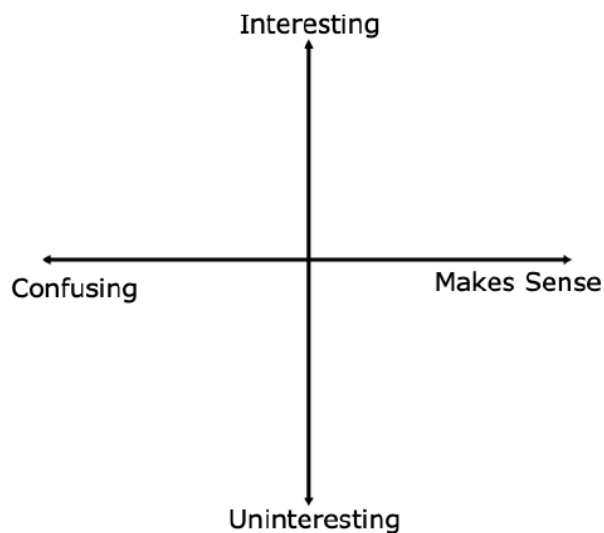
**Component Overview:** Students assess their understanding of describing and interpreting a histogram within a real-world context. Consider using this reflection to inform partner pairs or small groups for the next lesson, *Describing and Interpreting Histograms - Part 2*.



### 9.6.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate your level of understanding of today's learning target: "I can qualitatively describe and interpret a histogram within a real-world context."

*Answers will vary.*



Consider creating a chart to collect and organize data on student responses to the reflection. Put a mark beside each student according to their response. In this way, you will be able to gain better insight on the reflection by analyzing student data as a whole.

|           | Interesting | Uninteresting | Confusing | Makes Sense |
|-----------|-------------|---------------|-----------|-------------|
| Student 1 | X           |               | X         |             |
| Student 2 |             | X             | X         |             |